Priyadarshini College of Engineering, Nagpur

Department of Computer Technology

(Session 2025-2026)

The Smart Dairy Management System

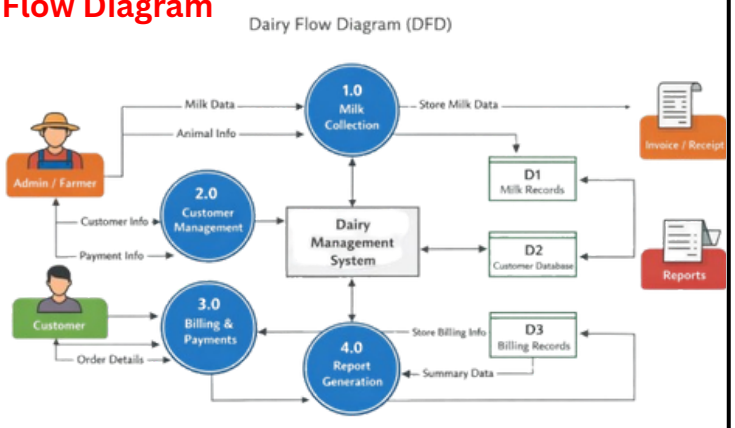
Dr.Nitesh Ghodichor (Computer Technology)

Name of Projectees: Sumit Karanjekar, Amit Taksande, Gaurav Chafle, Atharva Kurhe, Harsh Bhongade

Abstract: The Smart Dairy Management System is an AI-driven platform developed to automate and optimize dairy farm operations by integrating real-time monitoring and data analytics. Traditional dairy practices rely heavily on manual processes, which often result in inefficiencies and delayed decision-making. This system collects and analyzes data related to cattle health, milk production, and environmental conditions using Machine Learning algorithms to enable early disease detection, milk yield prediction, and performance evaluation. Built using web technologies such as HTML, CSS, JavaScript, PHP, MySQL, and Python-based ML models, the system provides a user-friendly interface for farmers to access insights and manage operations efficiently. The proposed solution enhances productivity, reduces manual effort, and supports data-driven decision-making, making it a scalable and effective approach for modern dairy farm management.

Introduction: The dairy industry is an important part of agriculture, but it still relies on traditional methods that reduce efficiency and productivity. Farmers face challenges like poor monitoring of cattle health, inconsistent milk production, and delayed disease detection. To overcome these issues, the Smart Dairy Management System uses Artificial Intelligence (AI) and Machine Learning (ML) to automate and improve dairy operations. The system provides real-time monitoring and data-driven insights, helping farmers make better decisions and increase overall productivity.

Flow Diagram



CO's

CO1: Acquire a sound technical knowledge for problem identification and formulation through the prior knowledge, literature, review and original ideas.

CO2: Use software engineering tools to analyse, design, implement, validate and maintain a project.

CO3: Develop solutions to the identified problems by applying and integrating the knowledge acquired throughout his/her undergraduate study and modern techniques.

CO4: Prepare and present a well-organised progress of a project in written and verbal form periodically.

CO5: Work in a team and communicate with superiors, peers and the community.

CO6: To publish and share their project works with outside world at national and international level.

POs

PO1: The Smart Dairy Management System is an AI-based platform designed to automate dairy farm operations and improve efficiency through data-driven insights.

PO2: Traditional dairy farming relies on manual monitoring, leading to inefficiencies, delayed decisions, and poor health tracking of cattle.

PO3: The system aims to monitor cattle health, predict diseases, and optimize milk production using Artificial Intelligence and Machine Learning.

PO4: It is developed using HTML, CSS, JavaScript, PHP, MySQL, and Python-based Machine Learning models.

PO5: The system includes features like real-time monitoring, disease prediction, milk yield forecasting, and performance evaluation.

PO8: The system reduces manual effort, increases productivity, enables early disease detection, and improves operational efficiency.

PO9: The project helps in developing technical knowledge, software engineering skills, teamwork, and communication abilities.

PO10: Future improvements include integration of IoT sensors, advanced ML models, automated systems, and multilingual support.

PO11: The system offers a scalable and efficient solution for modern dairy management, supporting sustainable and data-driven agriculture.

Program Specific Outcomes:

1. Anability to analyze a problem and identify its solution by applying knowledge of computing and fundamental concepts appropriate to the discipline. 2. An ability to design and develop a computerized systems using conventional and modern techniques, tools for solving real world engineering problems of varying complexity 3. An ability to employ the knowledge of Programme specific domains for professional growth and pursuing higher education to meet the current industrial needs

Publication:

1.Indian society for VLSI Education Ranchi 835217 (Jharkhand .India) Paper Id : NMIC26455_CMT455

Future Scope:

TheSmart Dairy Management System can be further improved by integrating IoT-based sensors for real-time monitoring of cattle health and environmental conditions. Advanced Machine Learning models can be used to increase the accuracy of disease prediction and milk yield forecasting. Additionally, adding features like automated feeding systems, alert notifications, and multilingual support can enhance usability and make the system more efficient and accessible for a wider range of users.

Application Area:

The Smart Dairy Management System can be applied in various areas such as dairy farm automation, cattle health monitoring, and milk production management. It helps in predicting diseases, optimizing milk yield, and improving overall farm efficiency using data analytics. The system is useful for small as well as large-scale dairy farms, supporting better decision-making and promoting modern smart agriculture practices.

Conclusion:

The Smart Dairy Management System provides an efficient and intelligent solution for modernizing dairy farm operations using Artificial Intelligence and
data analytics. It helps in monitoring cattle health, predicting diseases, and optimizing milk production, thereby reducing manual effort and improving productivity. The system supports data-driven decision-making and can be effectively implemented in both small and large-scale dairy farms, contributing to sustainable and smart agricultural practices.